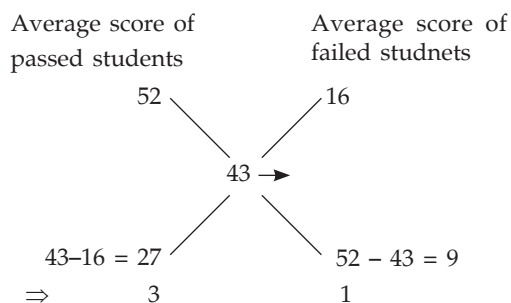


155. Total number of students in class = 60

Average score of passed students = 52

Average score of failed students = 16

By applying the rule of alligation,



$\therefore$ Number of students who failed in exam = $\frac{1}{4} \times 60 = 15$

156. The average of 11 results = 60

The total of 11 results = $60 \times 11 = 660$

Average of first six results = 58

Average of last six results = 63

Total of first six results = $58 \times 6 = 348$

Total of last six results = $63 \times 6 = 378$

$\therefore$ sixth results = total of first and last six results = total of 11 results

$$= (348 + 378) - 660$$

$$= 726 - 660 = 66$$

157. Average weights of 21 boys = 64 kg

Total weights of 21 boys = $64 \times 21 = 1344$ kg

The weight of the teacher was added then average increase by 1 kg

$\Rightarrow$ total weight of teachers and 21 boys = $65 \times 21 = 1430$ kg

Weight of teacher = $1430 - 1344 = 86$ kg

158. Average of 12 number = 15

Total of 12 number = $15 \times 12 = 180$

Average of first two number = 14

Total of first two number = $14 \times 2 = 28$

Total of remaining ten numbers = $180 - 28 = 152$

Required average of remaining ten number = $\frac{152}{10} = \frac{76}{5} = 15\frac{1}{5}$

159. Average expenditure of a man for the first five month

$$= ₹ 1200$$

Average expenditure of a man for the next seven month

$$= ₹ 1300$$

Total annual expenditure of man

$$= ₹ (5 \times 1200 + 7 \times 1300)$$

$$= ₹ (6000 + 9100)$$

$$= ₹ 15100$$

Man saves = ₹ 2900

His total annual income

$$= ₹ (15100 + 2900)$$

$$= ₹ 18000$$

$\therefore$ Average monthly income = $\frac{18000}{12} = ₹ 1500$

160. Let the number of female students be x

Let weight of female students = $57x$

Number of male students = 66

Total weight of male students = 65.9×66

Average weight of all the students 60.3 kg

Total weight of all the students = $60.3 (66 + x)$

According to given information

Then, $60.3 (66 + x) = 66 \times 65.9 + 57x$

$$60.3 \times 66 + 60.3x = 66 \times 65.9 + 57x$$

$$60.3x - 57x = 66(65.9 - 60.3)$$

$$\text{or, } 3.3x = 66(65.9 - 60.3)$$

$$\text{or, } 3.3x = 66 \times 5.6$$

$$\therefore x = \frac{66 \times 5.6}{3.3} = 2 \times 56 = 112$$

161. Total number of teachers = 10

Age of new teacher = 25 years

Age of the retired teacher

$$= (25 + 3 \times 10) \text{ years}$$

$$= 55 \text{ years}$$

162. Numbers are

$$x > y > z > p > q > r$$

According to the question,

Average of first five numbers = 30

Sum of first five number

$$= a + y + z + p + q = 5 \times 30 = 150 \quad \dots(i)$$

Average of last five number = 25

Sum of last five numbers

$$= y + z + p + q + r = 5 \times 25 = 125 \quad \dots(ii)$$

By equation (i) and (ii) $a - r = 150 - 125 = 25$

163. Average weight of A, B and C = 40 kgs

Total weights of A, B and C = $40 \times 3 = 120$ kgs

Weight of C = $(A + 24)$ kg and $C = (B - 3)$ kg

$$\therefore A + 24 = B - 3$$

$$\Rightarrow B = A + 27$$

Now $A + B + C = 120$

$$\Rightarrow A + A + 27 + A + 24 = 120$$

$$\Rightarrow 3A + 51 = 120$$

$$\Rightarrow A = \frac{69}{3} = 23 \text{ kg}$$

$$B = A + 27 = 23 + 27 = 50 \text{ kg}$$

$$C = 120 - 23 - 50 = 47 \text{ kg}$$

$$D = 47 - 15 = 32 \text{ kg}$$

$\therefore$ Required average weight of A, B, C and D

$$= \frac{23 + 50 + 47 + 32}{4}$$

$$= \frac{152}{4} = 38 \text{ kg}$$

164. Total weight of Arun + Suraj + Vinay = $68 \times 3 = 204$..(i)

Total weight of Raju + Pratap = $72 \times 2 = 144$ (ii)

Weight of Suraj = 78 kg